

Luís Rosa

+351 935 601 172 | luis.rosa17@hotmail.com | linkedin.com/in/luisapr1 | github.com/LuisAPR1 | luisapr1.tech

Agualva-Cacém, Lisbon, Portugal

EDUCATION

Instituto Superior Técnico

Master's in Computer Science

Lisbon, Portugal

Sep. 2025 – Present

University of Algarve

Bachelor's in Computer Science (Final Grade: 15/20)

Faro, Portugal

Sep. 2022 – Jun. 2025

PROFESSIONAL EXPERIENCE

Software and Web Developer (Trainee)

DigitSigma

Jul. 2022 – Aug. 2022

Loulé, Portugal

- Development of Web applications and Windows Forms (.NET).
- Database management, REST API integration, and development of full-stack features.
- Technologies used: C#, PHP, JavaScript, HTML, Bootstrap, Axios, and Ajax.

PROJECTS

ETL & OLAP Data Warehouse Pipeline | *SQL, Pentaho (PDI/PRD), Mondrian, MySQL, MDX, Saiku, Docker*

- Development of a complete Business Intelligence solution analysing ~1.3M UK real estate transaction records (HM Land Registry Price Paid Data) through a fully Dockerised pipeline for end-to-end reproducibility.
- Data Warehouse design (Kimball Star Schema) and dual ETL implementation in Pentaho Data Integration and idempotent pure SQL (recursive CTEs, set-based upserts) for cross-platform portability.
- OLAP cube modelling and KPI dashboards (Saiku, MDX) tracking metrics enabling data-driven decision-making.

ML Prediction Pipeline | *Python, Scikit-learn, ARIMA, LSTM*

- End-to-end Data Science pipeline applied to multiple real-world problems, including road accident severity prediction, flight cancellation detection, and traffic volume forecasting.
- Implementation and benchmarking of multiple classification models (Naïve Bayes, KNN, Decision Trees, Random Forest, Gradient Boosting, MLP) with a focus on realistic imbalanced class scenarios and hyperparameter tuning.
- Time series forecasting on high-frequency traffic data, comparing classical statistical methods (ARIMA) with deep learning approaches (LSTM) to support transportation infrastructure planning.

Permissioned Blockchain & BFT Consensus | *Java, Hyperledger Besu, Solidity*

- End-to-end development of DepChain, a highly dependable permissioned blockchain system, implementing both the consensus layer and a transaction processing engine to ensure Byzantine Fault Tolerance (BFT).
- Systematic implementation of the Basic HotStuff consensus algorithm over unreliable networks (UDP), building network abstractions such as authenticated perfect links and failure detectors to handle crash and malicious faults.
- Integration of EVM (Hyperledger Besu) for executing Solidity smart contracts, featuring a custom gas-fee mechanism and an ERC-20 token implementation redesigned to prevent Approval Frontrunning attacks.

Distributed Key-Value Store | *Python, FastAPI, Redis, RabbitMQ, CockroachDB, Docker, Kubernetes*

- Design and implementation of a fault-tolerant distributed key-value storage system, capable of managing data across multiple nodes with strong consistency and high availability guarantees.
- Multi-tier architecture combining a FastAPI service layer, Redis-based caching, RabbitMQ message queues for asynchronous coordination, and CockroachDB as the distributed persistent backend.
- Cloud-native deployment with Docker containers orchestrated by Kubernetes, including monitoring, health checks, and Horizontal Pod Autoscaling (HPA) to ensure resilience under node failures and dynamic load.

LANGUAGES

Portuguese (Native/C2), English (B2)